# IGCSE Chemistry homework · 2.1–2.3

CIE 0620 IGCSE Chemistry (Extended) · 42 marks · 45 minutes · Periodic Table allowed

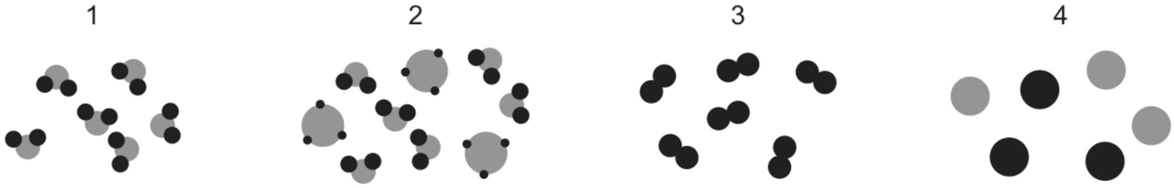
Topics: 2.1 elements, compounds and mixtures · 2.2 atomic structure and the Periodic Table · 2.3 isotopes (including relative atomic mass). Section A: 20 MCQ, 20 marks. Section B: B1 8 marks + B2 14 marks. Show working in structured questions.

| Section | Content | Marks |
|---|---|---|
| A | Multiple choice | 20 |
| B1 | 2021 ON Paper 42 Question 3 | 8 |
| B2 | 2024 ON Paper 41 Question 2 | 14 |
| | **Total** | **42** |

## Section A — Multiple-choice questions (20 marks)

**A1.** [1]

**2** The diagrams represent some elements, compounds and mixtures.



Which row describes the numbered substances?

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **A** | element | mixture of compounds | compound | mixture of elements |
| **B** | compound | mixture of compounds | element | mixture of elements |
| **C** | element | mixture of elements | compound | mixture of compounds |
| **D** | compound | mixture of elements | element | mixture of compounds |

**A2.** [1]

**2** Which row about elements, mixtures and compounds is correct?

| | metallic element | non-metallic element | mixture | compound |
|---|---|---|---|---|
| **A** | copper | methane | brass | sulfur |
| **B** | brass | sulfur | copper | methane |
| **C** | copper | sulfur | brass | methane |
| **D** | brass | methane | copper | sulfur |

**A3.** [1]

**4** Which row describes elements, compounds and mixtures?

| | elements | compounds | mixtures |
|---|---|---|---|
| **A** | all the atoms are the same type | contain different atoms chemically joined together | can be separated using physical processes |
| **B** | all the atoms are the same type | can be separated into new substances using chemical processes | contain different atoms chemically joined together |
| **C** | can be separated into new substances using chemical processes | all the atoms are the same type | contain different atoms chemically joined together |
| **D** | contain different atoms chemically joined together | all the atoms are the same type | can be separated using physical processes |

**A4.** [1]

**4** Four statements about atoms are listed.

1 The centre of an atom is positively charged.

2 Protons and electrons are located in the nucleus.

3 Protons and electrons have the same mass.

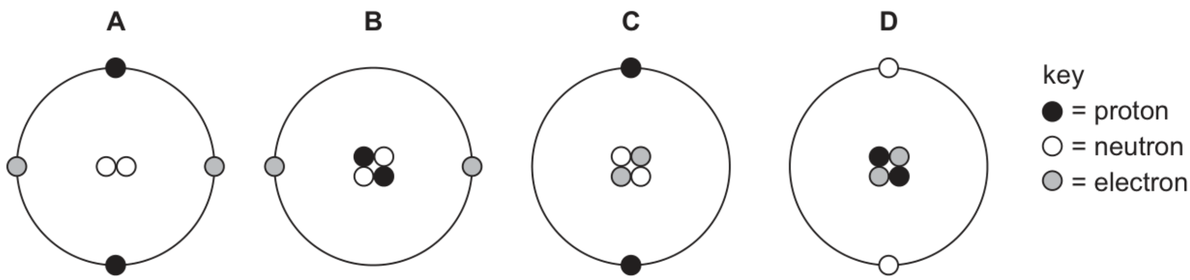
4 Most of the mass of an atom is in the nucleus.

Which statements are correct?

**A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

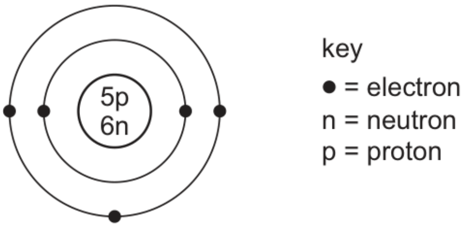
**A5.** [1]

**3** Which diagram represents one helium atom?



**A6.** [1]

**3** The structure of an atom of element X is shown.




What is element X?

**A** boron

**B** carbon

**C** sodium

**D** sulfur

**A7.** [1]

**3** The table shows information about four different particles.

| particle | nucleon number | number of protons | number of neutrons | number of electrons |
|---|---|---|---|---|
| Na | 23 | 11 | W | 11 |
| $Na^{+}$ | 23 | 11 | 12 | X |
| O | 16 | 8 | Y | 8 |
| $O^{2-}$ | 16 | 8 | 8 | Z |

What are the values of W, X, Y and Z?

| | W | X | Y | Z |
|---|---|---|---|---|
| **A** | 11 | 10 | 10 | 8 |
| **B** | 11 | 11 | 8 | 10 |
| **C** | 12 | 10 | 8 | 10 |
| **D** | 12 | 11 | 10 | 8 |

**A8.** [1]

**4** How many protons, neutrons and electrons are there in one atom of the isotope $^{27}_{13}Al$?

| | protons | neutrons | electrons |
|---|---|---|---|
| **A** | 13 | 13 | 13 |
| **B** | 13 | 14 | 13 |
| **C** | 14 | 13 | 13 |
| **D** | 14 | 14 | 13 |

**A9.** [1]

**4** Which particles have the electronic configuration 2,8,8?

1 an argon atom, Ar

2 an aluminium ion, $Al^{3+}$

3 a sodium ion, $Na^{+}$

4 a chloride ion, $Cl^{-}$

**A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

**A10.** [1]

**5** Nitrogen forms a nitride ion with the formula $N^{3-}$.

Which particle does **not** have the same electronic configuration as the nitride ion?

**A** $Al^{3+}$ **B** $Cl^{-}$ **C** $Na^{+}$ **D** $O^{2-}$

**A11.** [1]

**5** The electronic configurations of two elements are given.

element L: 2,8,8,1

element M: 2,8,4

Which row identifies the group number and the period number for element L and element M?

| | element L | | element M | |
|---|---|---|---|---|
| | group number | period number | group number | period number |
| **A** | I | 4 | IV | 3 |
| **B** | I | 4 | III | 4 |
| **C** | IV | 1 | III | 4 |
| **D** | IV | 1 | IV | 3 |

**A12.** [1]

**2** An atom of element X contains:

- 5 protons
- 6 neutrons
- 5 electrons.

Which statements about element X are correct?

1 X has an atomic number of 6.

2 X has a nucleon number of 11.

3 X is in Group II of the Periodic Table.

4 X is in the second period of the Periodic Table.

**A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

**A13.** [1]

**8** Some information about particles P, Q, R and S is shown.

| | nucleon number | number of neutrons | number of electrons |
|---|---|---|---|
| P | 12 | 6 | 6 |
| Q | 24 | 12 | 10 |
| R | 16 | 8 | 10 |
| S | 14 | 8 | 6 |

Which two particles are isotopes of the same element?

**A** P and Q **B** P and S **C** Q and R **D** R and S

**A14.** [1]

**3** The atomic structures of four particles, W, X, Y and Z, are shown.

| | electrons | neutrons | protons |
|---|---|---|---|
| W | 2 | 2 | 2 |
| X | 2 | 2 | 3 |
| Y | 2 | 3 | 2 |
| Z | 3 | 2 | 3 |

Which particles are isotopes of the same element?

**A** W and X **B** W and Y **C** X and Y **D** X and Z

**A15.** [1]

**3** Which statements about isotopes are correct?

1 Isotopes are atoms of different elements with the same number of protons.

2 Isotopes of the same element have the same chemical properties.

3 Isotopes are atoms with the same relative atomic mass.

4 Isotopes of the same element have the same electronic configuration.

**A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

**A16.** [1]

**5** Symbols representing four particles are shown.

$^{40}_{20}W$ $^{41}_{20}X^{2+}$ $^{37}_{18}Y$ $^{37}_{17}Z$

The letters are **not** the chemical symbols.

Which particles have the same number of neutrons?

**A** W and $X^{2+}$ **B** W and Z **C** $X^{2+}$ and Y **D** Y and Z

**A17.** [1]

**4** The symbols of four atoms are listed.

1 $^{12}_{6}C$

2 $^{14}_{6}C$

3 $^{14}_{7}N$

4 $^{29}_{14}Si$

Which atoms have more neutrons than protons?

**A** 1 and 2 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

**A18.** [1]

**4** The percentage abundances of three isotopes in a sample of neon are shown.

| isotope | percentage abundance / % |
|---|---|
| $^{20}_{10}Ne$ | 90.48 |
| $^{21}_{10}Ne$ | 0.27 |
| $^{22}_{10}Ne$ | 9.25 |

What is the relative atomic mass, $A_r$, of this sample of neon?

**A** 10.19 **B** 20.19 **C** 21.00 **D** 30.19

**A19.** [1]

**4** The only naturally occurring isotopes in a sample of europium are $^{151}Eu$ and $^{153}Eu$. This sample of europium has a relative atomic mass, $A_r$, of 152.

What is the percentage of $^{151}Eu$ present in this sample to three significant figures?

**A** 0.987% **B** 50.0% **C** 76.0% **D** 98.7%

**A20.** [1]

**6** A sample of copper has a relative atomic mass of 63.5.

What are the relative abundances of the isotopes in this sample of copper?

**A** 25% $^{63}Cu$ and 75% $^{65}Cu$

**B** 50% $^{63}Cu$ and 50% $^{65}Cu$

**C** 75% $^{63}Cu$ and 25% $^{65}Cu$

**D** 90% $^{63}Cu$ and 10% $^{65}Cu$

## Section B — Structured questions (22 marks)

**B1 · 2021 October/November Paper 42 Question 3 (8 marks)**

**B1(a)** [3]

**3** Atoms contain protons, neutrons and electrons.

**(a)** Complete the table to show the relative mass and the relative charge of a proton, a neutron and an electron.

| | relative mass | relative charge |
|---|---|---|
| proton | | |
| neutron | | |
| electron | $\frac{1}{1840}$ | |

[3]

**B1(b)** [5]

**(b)** The table shows the number of protons, neutrons and electrons in some atoms and ions.

Complete the table.

| atom or ion | number of protons | number of neutrons | number of electrons |
|---|---|---|---|
| $^{32}_{16}S$ | | | |
| $^{39}_{19}K^{+}$ | | | |
| | 35 | 44 | 36 |

[5]

[Total: 8]

**B2 · 2024 October/November Paper 41 Question 2 (14 marks)**

**B2(a)** [1]

**2** This question is about atomic structure and the Periodic Table.

**(a)** Define the term nucleon number.

.................................................................................................................................. [1]

**B2(b)** [1]

**(b)** State the connection between the number of occupied electron shells in an atom and the period number of that element.

.................................................................................................................................. [1]

**B2(c)** [2]

**(c)** Write the electronic configuration of the following atom and ion.

$^{28}_{14}Si$ ..............................................

$^{37}_{17}Cl^{-}$ ..........................................

[2]

**B2(d)** [5]

**(d)** Complete Table 2.1.

**Table 2.1**

| atom or ion | number of protons | number of neutrons | number of electrons |
|---|---|---|---|
| $^{23}_{11}Na$ | 11 | | |
| $^{19}_{9}F^{-}$ | 9 | 10 | |
| | 31 | 38 | 28 |

[5]

**B2(e)(i)** [2]

**(e)** A sample of thallium, T*l*, contains two isotopes, $^{203}Tl$ and $^{205}Tl$.

**(i)** Define the term isotopes.

..............................................................................................................................

.......................................................................................................................... [2]

**B2(e)(ii)** [2]

**(ii)** The relative abundance of $^{203}Tl : ^{205}Tl$ is in the ratio 3 : 7.

Calculate the relative atomic mass of thallium in the sample to **one** decimal place.

relative atomic mass = .............................. [2]

**B2(e)(iii)** [1]

**(iii)** Suggest why these two isotopes have identical chemical properties.

.......................................................................................................................... [1]

[Total: 14]